COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	08/15/2026	**DATE SUBMITTED:**	08/15/2026
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ACTIVITY NO. 6.1 LINEAR AND BINARY SEARCH

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a search tree in data structures?

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

3. What operations / implementations can be performed using binary and linear search operations?

4. What are the advantages in using binary search tree as data structure?

5. Give an example program using binary search and Linear search.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. What is a search tree in data structures?

- A search tree is a data structure that utilizes a tree format to systematically organize and store data, facilitating faster and easier searches for specific values. It is made up of nodes linked by branches, with the arrangement of each node adhering to specific guidelines based on the type of search tree. A prevalent example is the Binary Search Tree (BST), where smaller values are located on the left and larger values on the right of a node. Search trees also assist in other operations like insertion and deletion, rendering them effective for efficiently handling and accessing large datasets.

2. What are the Different types of search algorithm in data structures? Differentiate each type of search.

- The two prevalent kinds of search algorithms are Linear Search and Binary Search. Linear Search examines each item sequentially, beginning with the first element until the sought-after value is located. It is applicable to both sorted and unsorted data, although it may be slower with numerous elements. Binary Search functions by continually splitting a sorted array into two sections and determining which section holds the target value. It outperforms Linear Search for extensive datasets as it reduces the remaining elements by half with each iteration. Linear Search operates with a time complexity of O(n), whereas Binary Search functions with a time complexity of O(log n). Thus, Linear Search is easier to use and operates on unsorted data, whereas Binary Search is more effective but needs the data to be sorted.

3. What operations / implementations can be performed using binary and linear search operations?

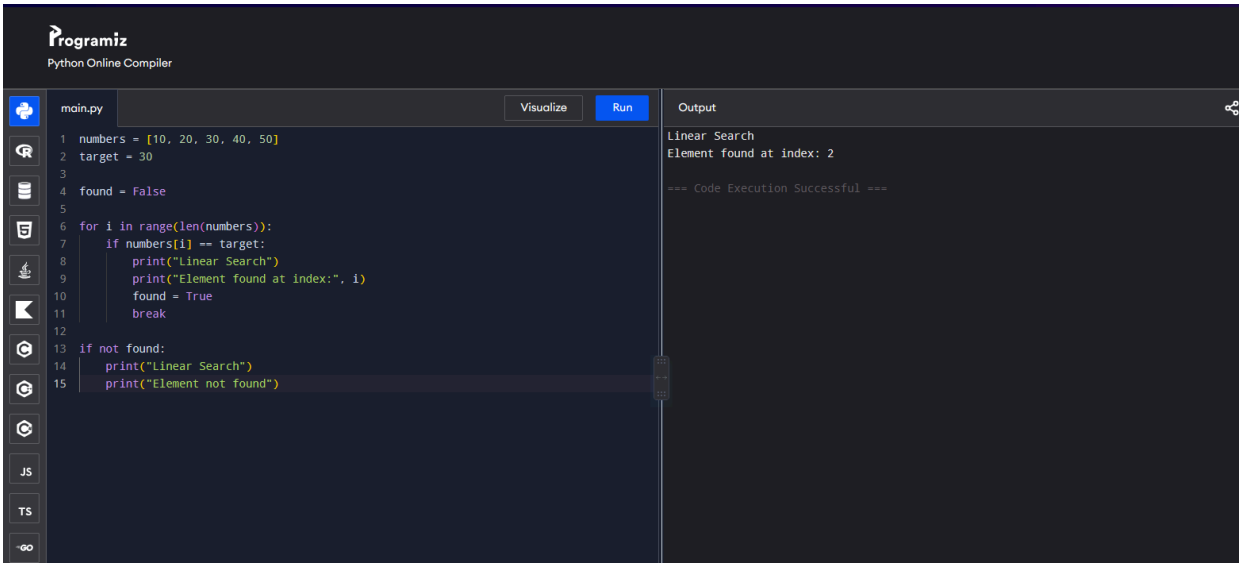
- Linear search can be used to find a specific element, check if an element exists, and find the minimum or maximum value in a list. Binary search can be used to find a specific element, locate its position, and find the first or last occurrence in a sorted list.

4. What are the advantages in using binary search tree as data structure?

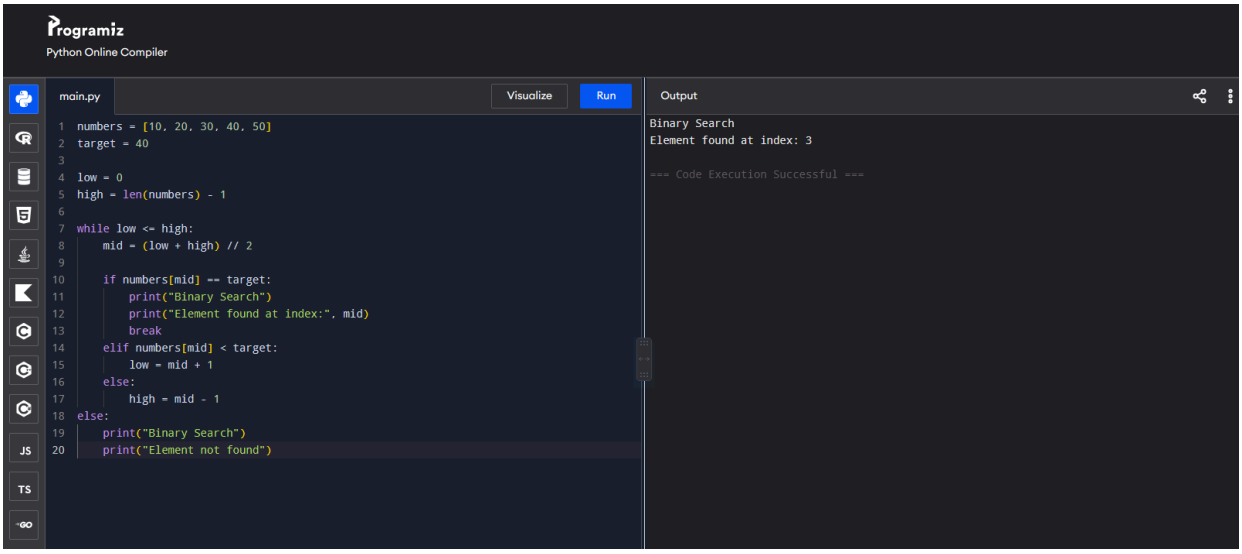
- The primary advantage of using a Binary Search Tree (BST) is that it bridges the gap between arrays and linked lists, offering efficient \(O(\log n)\) average-time complexity for search, insertion, and deletion operations. Unlike sorted arrays, it handles dynamic data modifications efficiently without needing to shift existing elements in memory.

5. Give an example program using binary search and Linear search.

LINEAR SEARCH



BINARY SEARCH



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **ReSearch**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **none**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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